

Kyung Hyun Lee

Clinical AI Researcher

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Summary

PhD student in Digital Health at SAIHST, Sungkyunkwan University. AI researcher working on clinical time-series deep learning, medical AI regulatory pathways, and respiratory digital health. Co-founder & CEO of BreathYou Co., Ltd.

Education

PhD Student	Sungkyunkwan University (SAIHST) , Digital Health • Advisor: Prof. Byung-Jae Lee • Research on Transformer-based trajectory models of respiratory health	Seoul, South Korea 2026 – present
Master of Science	Sungkyunkwan University (SAIHST) , Digital Health • Advisor: Prof. Mira Kang • M Lab. — Statistical analysis and predictive model development using medical big data	Seoul, South Korea 2019 – 2021
Bachelor of Science	Incheon National University , Electronic Engineering & Computational Neuroscience • Advisor: Prof. Sangwon Byun • Bioelectronics Lab. — Biosignal (ECG, EMG) data collection, signal analysis, and real-time prediction model development	Incheon, South Korea 2014 – 2019

Experience

University of Seoul , Lecturer Faculty appointment in the Department of Sport Science, teaching the undergraduate course Exercise and Digital Healthcare. • Designed and teach Exercise and Digital Healthcare, a fourth-year major elective (Fall 2026)	Seoul, South Korea 2026 – present 1 year
BreathYou Co., Ltd. , Co-founder & CEO Digital health startup focused on allergy and respiratory AI, developing DeepBreath CDSS and MediPipe. • Selected for 2025 Pre-startup Package • Incorporated on 5 August 2026 • DeepBreath CDSS: Clinical decision support for predicting pulmonary function decline trajectories • MediPipe: OCR/NLP pipeline for medical record processing	South Korea 2025 – present 1 year

AITRICS , Researcher (2024 – present) · Industrial Technical Personnel — Alternative Military Service (2022 – 2024) Medical data (RWD) cleaning, analysis pipeline construction, predictive model development, and external validation. Contributed to the cardiac arrest early-warning model that obtained MFDS (Korean FDA) approval as an AI medical device, and currently leading product development of a self-developed AKI prediction model toward MFDS approval. • Research design for clinical data analysis projects • Data extraction and cleaning from multi-center hospitals • Automated data analysis pipelines using Airflow • Deep learning-based clinical time-series prediction models • Clinical simulation-based model evaluation and survival analysis • Cardiac arrest prediction model — supported MFDS approval as an AI medical device • AKI prediction model — developed the model and currently leading productization toward MFDS approval	Seoul, South Korea 2022 – present 4 years
Exosystems , Biosignal AI Research Engineer Deep learning-based EMG modeling for Muscle Quality monitoring. • RNN, CRNN-based Muscle Quality time-series prediction model development • Clinical validation-oriented AI model evaluation	South Korea May 2022 – Aug 2022 4 months
Incheon National University , Researcher • EMG-based real-time hand gesture recognition algorithm R&D • Deep learning-based ECG modeling for autonomic nervous system monitoring	Incheon, South Korea Sept 2021 – May 2022 9 months
Samsung Medical Center & SAIHST , Researcher EHR data extraction and analysis from CDW; development of ML-based CDSS. • Data cleaning and health checkup database construction • Text mining of unstructured medical records • Statistical analysis and predictive/diagnostic model development	Seoul, South Korea Dec 2018 – Aug 2021 2 years 9 months

Teaching

University of Seoul, Department of Sport Science: Lecturer · Exercise and Digital Healthcare, fourth-year undergraduate elective (Fall 2026)

University of Seoul, Department of Sport Science: Invited seminar · Healthcare industry, two hours, hosted by Prof. Sae Young Jae (2024)

Incheon National University: Guest lecture · Student-Designed Convergence Major, one hour (2024)

Awards

Lab Startup Competition — Excellence Award Sungkyunkwan University, Start-up Support Foundation • University-wide competition among Sungkyunkwan University graduate-student laboratory startup clubs • MediPipe: on-premise medical data infrastructure software that turns in-hospital test results and clinical records into AI-ready data	2026
Consortium Lab-based Startup Camp — Excellence Award Sungkyunkwan University, Startup-Centered University program • Consortium competition among laboratory startup teams from universities across the Seoul metropolitan area • MediPipe: the data infrastructure that lets hospitals use AI	2026
Human Understanding AI Paper Competition — Top 7 & ETRI Director's Award ETRI, Ministry of Science and ICT • Multimodal Attention-based Korean emotion recognition model combining RoBERTa (Text) and Wav2vec 2.0 (Audio)	2022

OCR AI Training Data Hackathon — Excellence Award Ministry of Science and ICT, NIA AI-Assisted OCR Annotation System for pharmaceutical products (CRAFT-based)	2021
Medical Information Analysis Expert Advanced Course — Grand Prize Ministry of Health and Welfare, Asan Medical Center Real-time BIS prediction model using intraoperative EEG data (25-second lookahead)	2021
Medical AI Expert Training Course — Excellence Award Korea Human Resource Development Institute for Health & Welfare 30-day mortality and ICU admission prediction model using preoperative and intra-operative vital signs	2020
STAR Program Paper Presentation — Grand Prize DSP-IN-U Smart Device Talent Development Program EMG-based real-time hand gesture prediction model development	2018

Academic Service

Computers in Biology and Medicine: Peer reviewer · 4 manuscripts (2026)	
Information Processing & Management: Peer reviewer · 1 manuscript (2026)	
Frontiers in Medicine: Peer reviewer · 2 manuscripts (2026)	
Frontiers in Digital Health: Peer reviewer · 1 manuscript (2026)	
Frontiers in Network Physiology: Peer reviewer · 1 manuscript (2026)	
Frontiers in Psychiatry: Peer reviewer · 1 manuscript (2026)	
NeurIPS 2026 Workshop on Responsible Communication of Machine Learning Research in Biomedicine (RCMLR): Reviewer · 1 paper (2026)	

Certificates

Medical Information Analysis Expert Advanced Course Korea Human Resource Development Institute for Health & Welfare, Asan Medical Center	Aug 2021
Google Developer Machine Learning Boot Camp Google Developers	Jan 2021
Medical AI Expert Training Course Korea Human Resource Development Institute for Health & Welfare	Dec 2020

Skills

Programming & ML: Python, PyTorch, TensorFlow, Scikit-learn
Data & Database: SQL (PostgreSQL), Airflow, Data Pipeline
Statistics & Analysis: R, Survival Analysis, Clinical Statistics
Web Development: FastAPI, Flask, JavaScript, HTML/CSS, AWS

Languages

Korean Native speaker
English Professional working proficiency

References

Prof. Sangwon Byun Incheon National University, Department of Electronic Engineering; Bachelor's advisor (Bioelectronics Lab. · Scholar)

Prof. Mira Kang

Samsung Medical Center; Master's advisor at SAIHST, Sungkyunkwan University

Prof. Jin-Young Lee

Samsung Medical Center, Health Promotion Center

Prof. Byung-Jae Lee

Samsung Medical Center, Department of Allergy and Clinical Immunology; Doctoral advisor at SAIHST, Sungkyunkwan University

Prof. Sae Young Jae

University of Seoul, Department of Sport Science; research collaborator